AFFINE INTERSECTION CHARACTERIZATION OF QUASI-SEPARATED MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4358-4371, following the statement at 4341-4356.

Allowed standard prerequisites (editorial).

- Affine open covering of the fibre product. lemma-affine-covering-fibre-product (same chapter, cited at ~4360). V_j x_{U_i} V_j' cover X x_S X affinely.
- Quasi-compactness checked on affines. lemma-quasi-compact-affine (card L26389, cited at ~4364). Concludes qc of the diagonal pieces from finite affine unions.
- Open fibre products of affines. lemma-open-fibre-product (same chapter, used at ~4368-4369). U x_S V is affine open for the (1) => (2) direction.

Source reference. Original lemma 01JS: sources/stacks-schemes/pinned-original-source.tex, source lines 3448–3448. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01K4: sources/stacks-schemes/pinned-original-source.tex, source lines 3801–3801. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent:*

1. *The morphism f is quasi-separated.*
2. *For every pair of affine opens $U, V \subset X$ which map into a common affine open of S the intersection $U \cap V$ is a finite union of affine opens of X.*
3. *There exists an affine open covering $S = \bigcup_{i \in I} U_i$ and for each i an affine open covering $f^{-1}U_i = \bigcup_{j \in I_i} V_j$ such that for each i and each pair $j, j' \in I_i$ the intersection $V_j \cap V_{j'}$ is a finite union of affine opens of X.*

Proof. Let us prove that (3) implies (1). By Lemma 01JS the covering $X \times_S X = \bigcup_i \bigcup_{j,j'} V_j \times_{U_i} V_{j'}$ is an affine open covering of $X \times_S X$. Moreover, $\Delta_{X/S}^{-1}(V_j \times_{U_i} V_{j'}) = V_j \cap V_{j'}$. Hence the implication follows from Lemma 01K4.

The implication (1) $\Rightarrow$ (2) follows from the fact that under the hypotheses of (2) the fibre product $U \times_S V$ is an affine open of $X \times_S X$. The implication (2) $\Rightarrow$ (3) is trivial. $\square$

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.